

inter-related of course, you satisfy one, you automatically satisfy the other.

Of what use is education to a child with a rumbling tummy, of what use is electricity and Internet to a person who's dying from starvation? Ironically, the world was self-sufficient, in terms of food production almost 20 years ago; producing enough food to feed every person on this Earth the requisite 2100 daily calories. And even then, 800 million people suffered from severe malnourishment.

Think about that figure for a moment, that's more people than the entire population of Europe and represents over 70% of the population of India or China. Thankfully, this figure has fallen quite drastically to, but it still means that there are over 500 million starving people in the world, 200 million of whom reside in the African continent. Even more ironically, while food production in the world went up, the production in Africa fell and the number of starving more than doubled (from around 96 million to over 200 million).

This can be attributed to a variety of factors, but the most prominent of these are a lack of education to farmers (regarding something as elementary as crop rotation) to the inability to purchase, or lack of access to, high-yielding varieties

of crops, fertilizers, pesticides and other such absolute essentials.

Drawing the line

Many countries in Africa have lost more food crops to disease and drought and failed crops than they import. Organizations like the UN and the Gates foundation are doing a great deal to ensure production is improved, that farmers have access to genetically-modified crops (the high-yielding, disease resistant variety) and that they understand the intricacies of farming from a scientific standpoint, and all this will help for sure, at least in the short term. But the fact remains that we are producing more than we eat and if 30% of a country is starving, we're either eating way more than we should (which is partly true), but more significantly, we're wasting way more than we should.

The green revolution in the early 60s and 70s was essential to the survival of the human race. It was single-handedly responsible for more than doubling food production by dramatically improving yield and productivity per hectare. As dramatic as that was however, we're fast approaching a biological limit in terms of production. A quote from *The Economist* (<http://dgit.in/1AERYX>), we feel, best sums up the situation here, "*There must come a point where plants can no longer be sardined any*

closer together, stalk crushed against stalk. Similarly, at some point it may no longer be possible to persuade plants to put ever more energy into seeds. In animals biological limits are already clear: turkeys are so bloated that they can no longer walk; chickens grow so quickly that they suffer stress fractures."

So has the line already been drawn? Can we really produce no more?

A tech revolution

This might be hard to believe, but the messiah, the single most important factor that is transforming people's lives and ushering in a new era of prosperity and production, abolishing poverty and hunger, is the mobile phone.

Many of us believe that a famine is, by definition, a chronic shortage of food, and this is true. What most of us don't know is that even in times of famine, it's not always a hit to production that causes that shortage, but rather a shortfall in terms of purchasing power. This was most evident in the 1940s famine in Bangladesh that laid waste to millions.

How does the mobile phone fit in here? Taking the case of Africa, the mobile phone has completely transformed the concept of money and the traditional mechanism of conducting business transactions. The M-Pesa initiative (as it's known) is a mobile-phone based money transfer and microfinancing system (<http://dgit.in/1BiFeRz>) launched by Vodafone.

The service allows users to deposit and transfer money to and from accounts that are tied to their own mobile phones, via transactions initiated via SMS. This simple system has completely revolutionized money in Africa because it provided access to quick and easy credit (micro-finance), giving farmers the purchasing power and means to provide for their farms and, in-effect, their livelihood.

Oh, and Apple Pay (and other such services), please take note: M-Pesa is the most viable mobile-based money transfer system that exists today. Despite a number of audits by concerned banks (threatened by the success of M-Pesa) the service has proven to be robust and mostly fool proof. The service has grown well beyond a simple transaction portal and is now also being used in countries like Afghanistan to pay governmental salaries, and has inadvertently been responsible for a mas-

ENERGY TRANSFORMATION

Motive power is an essential to the success of civilization. What's the use of farming and other forms of production if that food and produce can't be moved distributed to the population as a whole? We take motive power for granted today of course, in a reasonably civilized part of the world that is, but even a city like Mubai or Delhi will not survive longer than a week if the transport system breaks down.

That said, we're so heavily reliant on transportation and motive power that a long term concern for the environment pales in comparison to the needs of immediate survival. Sure, a 100 years from now, if we don't change our way of life, the human race might die out; but right now, tomorrow, there'll be hell to pay if there's no milk or eggs in every household in the morning. Our dependence on fossil fuels will be our undoing, but our dependence on motive power will also be our undoing. We can't do anything about the motive power bit, expansion is a given, but we can do something about the fossil fuels. In our opinion, the biggest players in this space have to be Google and Elon Musk.

Why these two? Well, Elon Musk for working on producing a viable and sustainable electronic alternative to the internal combustion engine and crude oil and Google for working on their driverless car program. Both of these are a very big deal as far as the future of the world is concerned, especially if there were to exist some sort of synergy between the two plans for the future of transport. Imagine a world that's powered by a whole fleet of self-driving vehicles (including transport trucks) and electricity, automatically charging en route. The savings in terms of noise pollution alone will be worth the effort and the money. It's a future that offers a slender hope for our survival.